

Research papers

Employing stable isotopes to reveal temporal trajectories of water travelling through the soil–plant–atmosphere continuum

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ABSTRACT

The spatiotemporal patterns of water travelling through the soil–plant–atmosphere continuum (SPAC) have received much attention due to the frequency of extreme weather and water scarcity. However, the interconnections and transboundary transport of specific compartments within the hydrologic cycle are poorly understood. We portrayed the propagation paths of water isotope signals by analyzing isotopes of precipitation, soil water, and xylem water. The isotope sine curves analysis method was improved to quantify water transfer efficiency, mean transmission time from precipitation to soil (MTT₁), mean transmission time from soil to trees (MTT₂), mean residence time within soil (MRT_{SW}), and mean residence time within tree xylems (MRT_{XY}). The temporal trajectory of water traveling through SPAC was depicted, and its influencing factors and intrinsic mechanisms were explored. Our results showed that (1) water isotopes were blocked when crossing water pools (precipitation, soil water, and xylem water) and the transfer efficiency was progressively weaker (85.5 %–13.5 %). (2) The MTT₁ was significantly and positively correlated with soil porosity (Spearman correlation coefficient, 0.551). Its spatiotemporal pattern (0.8–15.9d) indicated that the transport and mixing of soil water involved two modes: displacement flow and bypass flow. The MTT₂ (–13.2d–4.3d) of *Platycadus orientalis* was in general smaller than that of *Quercus variabilis* (–4.7d–11.5d). (3) The MRT (35.2d–343.8d) was influenced by a combination of soil physicochemical properties, root distribution, and tree morphological traits. Our study shows that connectivity between pools is better reflected with transfer efficiency. Comparing MTT and MRT of *P. orientalis* and *Q. variabilis*, tree hydrodynamic processes were quantified, and *P. orientalis* was more sensitive to seasonal moisture variability. Additionally, our study improved the understanding of the “black box” within the hydrological interface of plant–soil interactions.

1. Introduction

With the frequent occurrence of extreme weather and increasing inequality in water allocation (Miguez-Macho and Fan, 2021), the process of water transport in SPAC has received much attention, and studies on water transit time and flow paths are crucial for solving many eco-hydrological questions.

The water transport process in SPAC exhibits a certain time lag (Geris et al., 2017). Precipitation infiltrating the soil does not immediately and completely mix with “old water” (precedent soil water) (Tetzlaff, et al., 2021). When water reaches the root zone, it is not instantly absorbed by plant roots (Luo et al., 2021); similarly, the water absorbed by plant

roots is not immediately used for leaf transpiration (Mennekes et al., 2021). In other words, the temporal trajectories of water travel reflect hydrological processes (Sprenger et al., 2016). The assessment of connectivity between water compartments is particularly important in order to explore the internal causes of time lag. Hydrological connectivity reflects the transboundary and feedback effects of water transport between compartments (Sprenger et al., 2019b). Actually, hydrological connectivity can also be considered as water transfer efficiency (Goldsmith et al., 2019). Exploring the response of soil water content to precipitation is a simpler example of assessing precipitation–soil water connectivity. More often, water stable isotopes are used as a tool to assess hydrological connectivity (Jia et al., 2022; Renée Brooks et al.,

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2010).

Water stable isotopes have become an important tool for tracing water transport pathways, as well as revealing patterns of water mixing and transport between individual compartments of SPAC (Sprenger et al., 2019b). The integration of stable water isotopes with hydrological models can simulate precipitation infiltration processes in the soil (Huang et al., 2015; Sprenger et al., 2015), assess the connectivity between plant water uptake and potential water sources (Jia et al., 2022), assess the spatial (Liu et al., 2019) and temporal (Allen et al., 2019; Martin et al., 2018) sources of plant water use, and assess the temporal offset between precipitation and plant water uptake (Juhlke et al., 2021). Sprenger et al., 2018 highlighted the significant seasonal variations in precipitation isotopes, and these seasonal fluctuations are transmitted as water travels through SPAC. The waveforms and frequencies of isotope signals are affected by different media (von Freyberg and Grossiord, 2020), thereby isotope signals can trace the complex time trajectory of water movement.

Water transmission time (WTT) in soil can be considered as the time required for precipitation to infiltrate the soil at a given depth (Brinkmann et al., 2018). And the period of time during which water remains in a given soil layer is referred to as water residence time (WRT) in soil. Because soil water at a specific depth originates from multiple precipitation events, WTT and WRT in soil is not a single value. Instead, they represent the average of all movement times of the soil water at a particular depth (Brinkmann, 2018; Luo et al., 2021). Therefore, the mean transmission time (MTT) and mean residence time (MRT) serve as valuable indicators reflecting the capacity of soil to hold and discharge water. Many studies have applied relevant isotope hydrological modeling to quantify MRT of precipitation in soil and explore its relationship with environmental and plant physiological factors. Sprenger et al. (2019a) indicated that MRT varies with soil depth and soil pores. The variations in both the isotope signature in precipitation and soil water were periodic, with a trend close to a sine function. Through model fitting and calculation, by comparing the amplitudes and displacements between the fitted curves of precipitation and soil water, the MRT and MTT can be inferred (Kabeya et al., 2007; Reddy et al., 2006; Rodgers et al., 2005).

For WTT from soil to plant, considerable research has focused on exploring the time lag of water uptake by plants (Wan, 2023; Zhao and Wang, 2021; Mennekes et al., 2021). The irregular distribution of plant roots prevents plants from utilizing potential water sources in chronological order. Thus, the interaction between plant root distribution and soil water infiltration patterns causes plants to utilize precipitation from previous seasons rather than recent precipitation (Allen et al., 2019); this creates a time lag in water uptake by plants. For example, Rempe, 2018 found that in arid or Mediterranean climates, plants primarily utilize precipitation from past seasons. In addition, the transmission time for water transport within trees should not be ignored. Gaines et al. (2016), by combining stable isotopes and sap flows to estimate the time lag from root water uptake to transpiration, found that peak concentrations of monitored tracers lagged one day to one week from the time of input. Ultimately, they assessed the lag times of four tree species to be 5–22 days. The MTT of groundwater recharge was two to seven times faster than that of transpired water. Transpired water age demonstrates a strong dependency on species and is related to the differences between midday leaf water potential and soil matric potential (Evaristo et al., 2019). In a tropical rainforest system, the transpired water age varies from hours to 12 days (Correa et al., 2020). Furthermore, water transient storage and mixing dynamics within trees lead to difficulties in defining and quantifying the MRT of water within trees. Our understanding of the extent to which old water stored in trees mixes with new water taken up by the roots is limited.

Identifying and quantifying WTT and WRT in SPAC helps to determine the pattern, timing, and mechanism for the connection of water of different ages in the water cycle (Sprenger et al., 2019b). Previous studies of water stable isotopes in SPAC have traditionally relied on data

collected over relatively brief periods (Sprenger et al., 2017). Some studies have confused WTT with WRT (Ma et al., 2019; Wu et al., 2022). In other studies, only the time trajectory of one water component was studied, either focusing on soil water or plant water alone, without linking the three pools of precipitation, soil water, and xylem water, so the understanding of water movement pathways was not sufficiently clear. Studies at the ecosystem scale with spatiotemporal heterogeneity of water are lacking. A comprehensive and systematic understanding of the water travel process and water hysteresis mechanism in SPAC is of scientific significance for decreasing the uncertainty of hydrological and climatic process models and assisting in decision-making for future vegetation planting measures.

We investigated the spatiotemporal variations in stable isotopes in precipitation, soil water at different depths, and plant xylem water in a temperate mountainous forest over a three-year period from 2015 to 2017. The water travel processes in SPAC were illustrated in combination with relevant environmental variables. Simultaneously, the seasonal isotope sine curve analysis method of Allen et al. (2018) was improved to meet local climatic and other conditions to explore the water transport pathways, transfer efficiency, MTT, and MRT in different water pools. This method provided new insights into water transport processes and residence times, and aimed to improve understanding of the “black box” within the hydrological interface of plant-soil interactions. The main objectives of this study are as follows:

- (1) to describe the isotopic variations within water pools and the connectivity between water pools;
- (2) to quantify the spatiotemporal heterogeneity of MTT and MRT in SPAC hydrological processes using an innovative isotopic signal method; and
- (3) to explore the influencing factors of MTT and MRT and underlying mechanisms.

2. Materials and methods

2.1. Study site

The study site is located within the National Forest Ecosystem Research Station (40°03'27"N, 116°05'31"E), which covers an area of about 16 km² in Beijing, China. The region is characterized by a semi-humid, semi-arid, continental monsoon climate. For the past two decades, the mean annual temperature of the site has been 11.6°C, with the annual precipitation averaging at 660 mm, with approximately 70 % falling in June–September. The estimated annual potential evapotranspiration is approximately 1100 mm (Jia et al., 2022). The soil type is mainly brown soil with a high clay content at shallow depths. The deeper soils are loamy in texture and contain significant amounts of gravel (Wenping, et al., 2021). We selected a mixed forest comprising *Platycladus orientalis* (conifers) and *Quercus variabilis* (broadleaf trees).

2.2. Meteorology, soil moisture, and soil property measurements

Outside the sample plots, the HOBO weather station (Onset, USA) monitored meteorological data including temperature, relative humidity, solar radiation, and precipitation. Soil volumetric water content (VWC) (m³·m⁻³) was determined by soil moisture probes (5TE, METER, USA) placed in the sample plots at depths ranging from 0 to 100 cm, with depth intervals of 20 cm and monitoring intervals of 10 min. The recharge of soil by seasonal precipitation was studied by comparing the change in soil VWC before and after precipitation. The change in soil moisture, denoted ΔSWC , is calculated according to the difference between SWC of two consecutive days (Liu et al., 2022b). The larger the ΔSWC , the greater the recharge effect of precipitation. Meanwhile, the difference in ΔSWC along each soil depth indicates that precipitation-soil connectivity varies by soil depth (Dai et al., 2020). We also measured soil property indicators including soil porosity, soil separation

composition, and soil bulk density. The measurement methods are described by Liu et al., 2019.

2.3. Characterization of root distribution

In this study, the tree root system was sampled during the growing seasons during the study period to obtain tree root distribution data. We selected three sample plants for each of the two tree species and extracted the root system at 10 cm intervals using a root drill (7 cm internal diameter, 10 cm height) at a distance of 50 cm from the trunk of each tree. After taking the samples to the laboratory, we used sieves to categorize the samples according to the classification system of diameter (d) less than 2.0 mm, 2.0–5.0 mm and greater than 5.0 mm (Liu et al., 2019). The sieved roots were then dried in an oven at 105 °C until the constant weight and dry weight of the roots (g) (root biomass) was measured. To clarify the distribution characteristics of root biomass in the vertical profile, we calculated the root biomass ratio (%), which is the weight of each root diameter category in the soil profile as a percentage of the total root biomass. To understand the biological characteristics of each sampled tree, we included DBH (diameter at breast height) and height in the measurements.

2.4. Isotope sampling and analysis

Precipitation samples were obtained using self-designed rain barrels with fractionation avoidance in open areas near weather stations (Jia et al., 2022). Using soil augers, we collected soil samples in layers within a 50 cm radius from each tree trunk, consistent with root sampling rules. We collected emboldened branches from the mid-canopy layer of each tree, removed the epidermis and phloem, and stored them until processing. Three replicates were performed for each sample, with a sampling frequency of 3 times per month. All precipitation, soil, and branch xylem samples collected were stored in 50 ml vials, sealed with Parafilm M®, and refrigerated at –20°C until water isotope measurements were undertaken.

Sample measurements were performed in the Laboratory of Ecohydrological Processes and Mechanisms (Beijing Forestry University, China). First, water was extracted from the branch and soil samples using an automatic cryogenic vacuum extraction system (LI-2100, LICA, China), and the isotope signatures of all extracted water samples were measured with a Liquid Water Isotope Analyzer (DLT-100, LGR, USA). The $\delta^2\text{H}$ and $\delta^{18}\text{O}$ accuracies of this instrument were $\pm 0.35\%$ and $\pm 0.18\%$, respectively. The isotope signature was expressed as:

$$\delta^2\text{H or } \delta^{18}\text{O} = \left(\frac{R_{\text{sample}}}{R_{\text{standard}}} - 1 \right) \times 1000\% \quad (1)$$

where R_{sample} was the heavy and light isotope ratio of the corresponding element in the sample, and R_{standard} was the heavy and light isotope ratio of the element in the international standard, referred to as the Vienna Standard Mean Ocean Water (VSMOW).

2.5. Statistical analysis

The linear relationship between $\delta^2\text{H}$ and $\delta^{18}\text{O}$ in precipitation is defined as the MWL (meteoric water line) (Craig, 1961), and the linear distance between isotope signature in different pools and the LMWL (local meteoric water line) is defined as line-conditioned excess (lc-excess), which can be used to quantify the degree of isotopic deviation between different water pools and local precipitation (Hervé-Fernández et al., 2016). The specific calculation process was as follows.

$$\text{lc-excess} = \delta^2\text{H} - a \times \delta^{18}\text{O} - b \quad (2)$$

where a and b were the slope and intercept of the LMWL, respectively, and $\delta^2\text{H}$ and $\delta^{18}\text{O}$ were the stable isotope composition in the water samples.

The seasonal isotope sine curve analysis method proposed by Allen et al. (2018) was improved. We fitted sine curves to the stable isotope time series of precipitation, stratified soil water and plant xylem water, and calculated MTT and MRT based on the fitted sine parameters. We used an additive model of sine and cosine functions set to a fixed one-year period. This model was adjusted using two amplitude parameters, a_1 and a_2 , and a constant offset parameter, *offset*. The specific calculation process was as follows:

$$\delta^{18}\text{O or } \delta^2\text{H}(t) = a_1 \times \cos(2\pi t) + a_2 \times \sin(2\pi t) + \text{offset} \quad (3)$$

Eq.3 was then transformed into a sine function:

$$\delta^{18}\text{O or } \delta^2\text{H}(t) = \text{amplitude} \times \sin(2\pi t - \varphi) + \text{offset} \quad (4)$$

where t (time) was defined in decimal years and amplitude and φ (phase) were calculated as follows:

$$\text{amplitude} = \sqrt{a_1^2 + a_2^2} \quad (5)$$

$$\varphi(\text{radians}) = -\text{atan2}(a_1, a_2), \text{ or,}$$

$$\text{aspeakdayofyear} = 365 \frac{\text{days}}{\text{year}} \left(\frac{1}{4} + \frac{\varphi}{2\pi} \right) \quad (6)$$

Sine curves were fitted to each of the three pools of precipitation, soil water, and xylem water, MTT was characterized by calculating the phase difference between the sine curves of the different water pools. The specific calculations were as follows:

$$\text{MTT}_1 = c^{-1}(\varphi_{\text{sw}} - \varphi_{\text{p}}) \quad (7)$$

$$\text{MTT}_2 = c^{-1}(\varphi_{\text{xy}} - \varphi_{\text{sw}}) \quad (8)$$

where MTT_1 represented the phase difference between the isotope sine curves of soil water and precipitation, and MTT_2 represented the phase difference between the isotope sine curves of plant xylem water and soil water; where φ_{p} , φ_{sw} , φ_{xy} represented the phases of precipitation, soil water, and xylem water sine curves, respectively; and c was the angular frequency constant ($2\pi/365$), dimensionless.

The mean residence time (MRT) represented the mean time during which water remained in a particular water pool from the time of entry. The specific calculations were as follows:

$$\tau = c^{-1} \sqrt{f^{-2} - 1} \quad (9)$$

where τ was MRT (day) and f was the damping factor (DF), denoted as (f) which represented the transfer efficiency of isotope signal waves from the output pool to the input pool. The specific calculations were as follows:

$$f = \frac{A_n}{A_m} \quad (10)$$

where A_n was the amplitude of the isotope output function and A_m was the amplitude of the isotope input function. When precipitation isotopes were used as the input function, soil-water isotopes were used as the output function. When soil water isotopes were used as the input function, xylem water isotopes were used as the output function.

Because the $\delta^2\text{H}$ -based sine curves and $\delta^{18}\text{O}$ -based sine curves of the pools were basically the same in terms of variation characteristics, and the fit of the former was better than that of the latter, only $\delta^2\text{H}$ was used for the calculations.

We used one-way ANOVA to test for significant differences among samples that passed the homogeneity of variance test. Before conducting the ANOVA, we verified the normal distribution and homogeneity of the variance of the data under consideration. Owing to unequal sample sizes across different seasons, we applied the Kruskal-Wallis (KW) test to identify any significant variations in the isotope signatures among

seasons. Spearman's correlation was used to detect the correlation between root biomass distribution, volumetric soil water content (SWC), soil porosity, soil separation composition, and soil bulk density with MTT and MRT. The fitting of isotope sine curves and statistical analyses were conducted using R software (see Fig. 1).

3. Results

3.1. Environmental conditions

Meteorological indicators in the study area were continuously monitored and analyzed (Fig. 2). In 2015, 2016, and 2017, the recorded precipitation levels were 580, 649, and 309 mm, respectively, with obvious interannual differences. The interannual precipitation distribution showed significant seasonal variation. The total precipitation during the rainy season (June–September) in 2015 was 458.2 mm, accounting for 79 % of the annual precipitation; the total precipitation during the rainy season (July–October) in 2016 was 592.8 mm, accounting for 91.2 % of the annual precipitation; and the total precipitation during the rainy season (June–August) in 2017 was 233.4 mm, accounting for 75.5 % of the annual precipitation. Over the three years, the highest monthly precipitation occurred in July, July, and August, respectively. Precipitation in January and December was 0 mm for all three years. Solar radiation in 2017 ($84.19 \pm 2.43 \text{ W m}^{-2}$) was significantly lower than the levels recorded in 2015 ($130.97 \pm 4.07 \text{ W m}^{-2}$) and 2016 ($75.83 \pm 5.07 \text{ W m}^{-2}$), as determined by one-way ANOVA ($p < 0.05$).

The SWC showed strong seasonality, which was consistent with precipitation inputs in different seasons (Fig. 2b). During the rainy season, soil moisture reached near-saturation levels, particularly in the top layer. At the same time, there was little difference in SWC among soil layers ($p > 0.05$, two-way ANOVA). The SWC of the 0–20 cm soil layer was significantly lower than that of the other soil layers during the dry season, except in 2015 ($p < 0.05$, two-way ANOVA). The average SWC of all the soil layers peaked in July during the observation period.

The results showed that from 2015 to 2017, there were 44, 36, and 21 precipitation events, respectively (Fig. 3). Overall, it appeared there was a significant positive correlation between ΔSWC of shallow soils and precipitation ($p < 0.01$), which meant that the soil layers with significant changes in response to precipitation were mainly concentrated in the range of 0–40 cm, indicating that shallow soils were very sensitive to the response to precipitation. This was more consistent with the characteristics of the changes in SWC at the corresponding depths indicated in Fig. 2b. Based on the intercept of the regression line equation in Fig. 3, it was estimated that, on rainless days, shallow soil loses moisture by 0.802 % on average. Under heavy precipitation, the depth of infiltration increased, and the opportunity for mixing precipitation with deep soil

water increased; therefore, the degree of mixing was enhanced.

The root biomass of *P. orientalis* and *Q. variabilis* showed different vertical distributions in the soil profile (Fig. 4). *Platycladus orientalis* had a relatively rich surface root system, with 45.0 % of the total root biomass in the 0–20 cm soil layer, which was significantly higher than the 2.7 % in the 80–100 cm soil layer ($p < 0.01$). In contrast to the root biomass distribution pattern of *P. orientalis*, *Q. variabilis* had a more pronounced primary root system. The root biomass of *Q. variabilis* with a diameter greater than 5 mm was significantly higher across the soil profile than those with diameters less than 2 mm and diameters ranging between 2 and 5 mm ($p < 0.05$). The root biomass of *Q. variabilis* with a diameter greater than 5 mm accounted for 79.2 % of the total biomass. In contrast, the root biomass of *Q. variabilis* with diameters less than 2 mm and ranging between 2 and 5 mm, accounted for less than 1 % in the 60–80 cm soil layer.

3.2. Seasonal variation of isotopes within water pools

Based on the collected samples, we analyzed the isotopic variations within different water pools. Since the seasonal variations in the stable isotopes of hydrogen and oxygen are essentially the same, we only discuss the characterization of the $\delta^2\text{H}$ variations in this paper.

Seasonal variations in precipitation isotope signatures were found in the research area (Fig. 5), with a significant negative correlation with rainfall during the rainy season ($p < 0.05$, one-way ANOVA) and no significant linear relationship during the dry season ($p > 0.05$, one-way ANOVA). The precipitation isotope signature became progressively depleted after the start of the rainy season and progressively enriched after the end of the rainy season. During 2015–2017, the most depleted isotope signature appeared in August (-90.15 ‰), August (-96.91 ‰), and September (-118.63 ‰), respectively, while the most enriched isotope signature appeared in February (-35.93 ‰), February (-39.25 ‰), and March (-21.50 ‰), respectively.

During the study period, the soil water isotope signatures in each layer were more enriched in the dry season than in the rainy season ($p < 0.05$, KW test) (Fig. 6), indicating that the isotopic fractionation effects of soil water in each layer were more pronounced in the dry season than in the rainy season. Although the isotopic composition of soil water was significantly different from that of precipitation ($p < 0.05$, one-way ANOVA), it still mirrored the temporal trends of the precipitation's isotopic signature, except that the magnitude of change was small and there was a time lag. For example, the most enriched monthly isotope signature for 20–80 cm soil water appeared in February (-31.07 ‰ – -67.36 ‰), and the most depleted monthly isotope signature appeared in September (-74.80 ‰ – -99.80 ‰). There was a time offset from the month in which the peak precipitation isotope signature appeared. In contrast, seasonal variations in isotope signatures of deep soil water

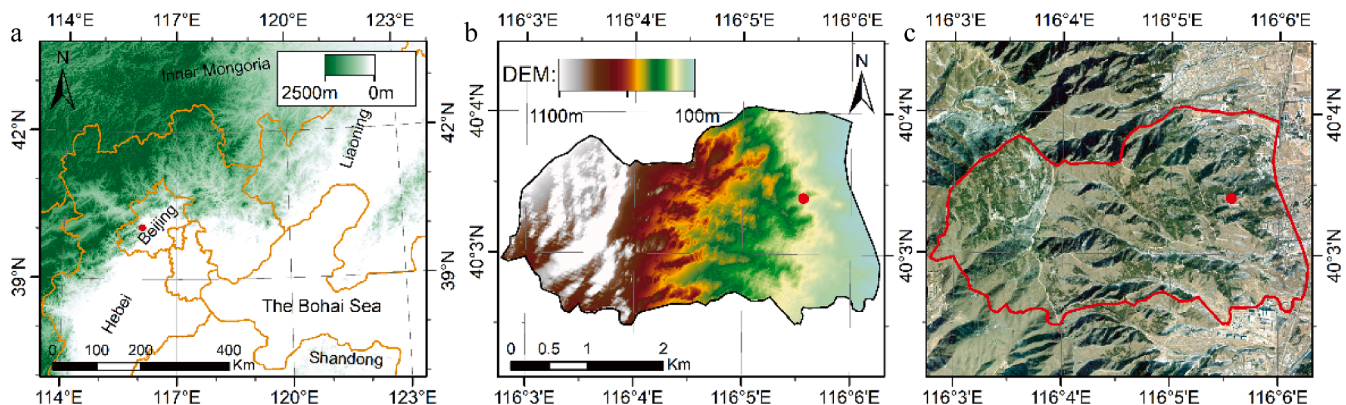


Fig. 1. Geographic information of the study area. Subfigure (a), (b), and (c) represents the location, elevation and satellite image of the study area, respectively. The red dot in (b) and (c) represents sample plots.

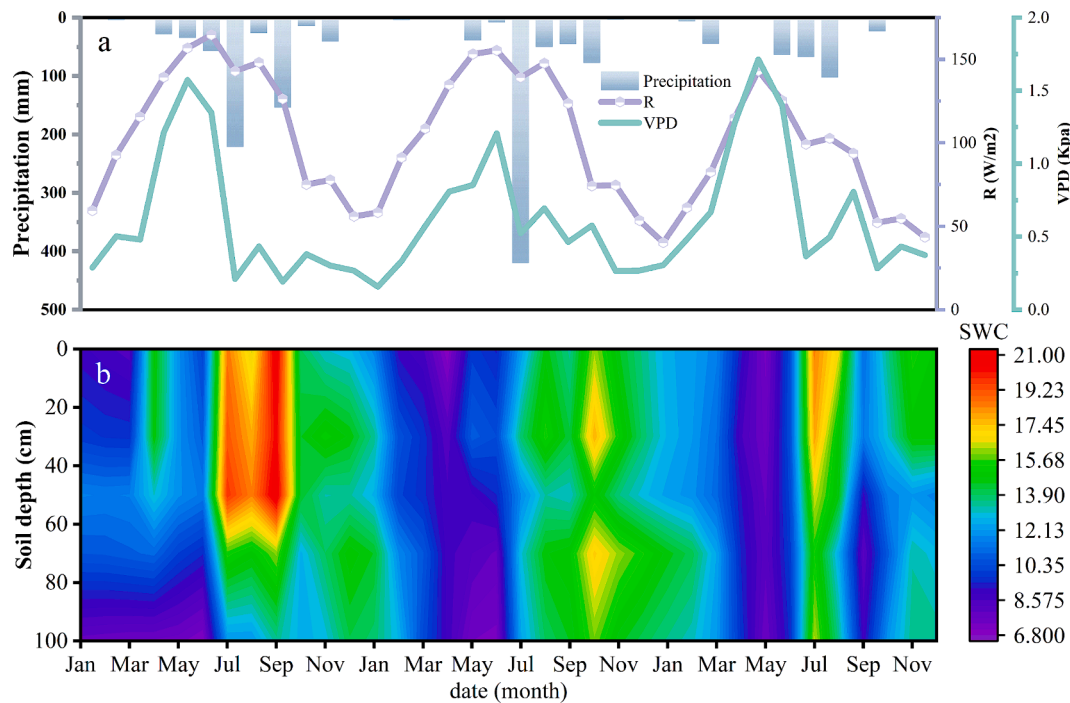


Fig. 2. Variation of (a) precipitation, R (solar radiation), VPD (vapor pressure deficit), (b) SWC at different soil depths throughout the study period.

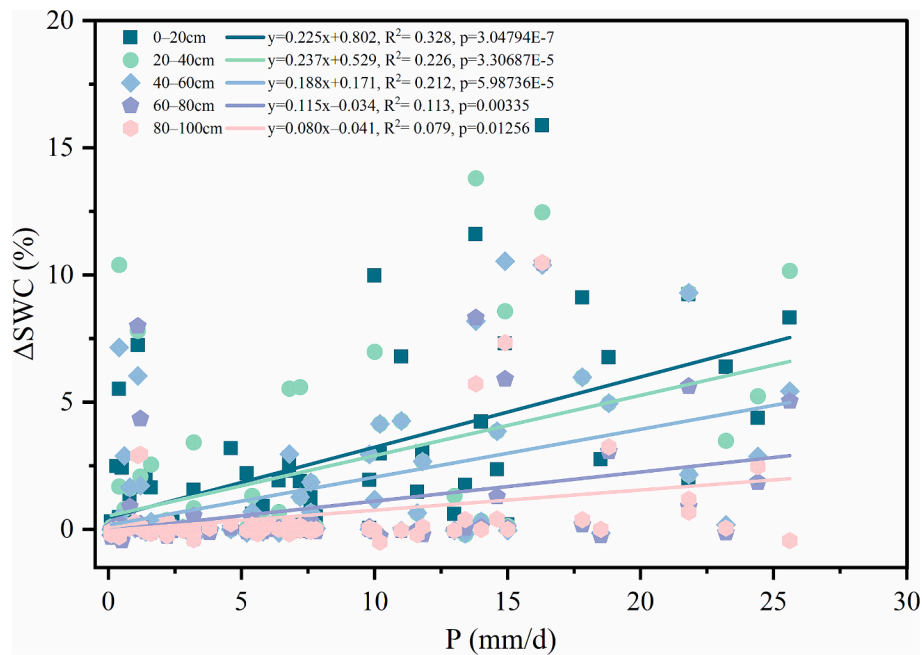


Fig. 3. Scatter plot showing the correlation between the variation in SWC (ΔSWC) and the daily precipitation (P) over two consecutive days. The lines in the figure are the fitted curves of precipitation and ΔSWC in different soil layers, respectively.

(80–100 cm) were weak, and the months in which the peaks and troughs of isotope signatures appeared differed each year.

Platycadus orientalis xylem isotope composition (Fig. 7) showed significant differences between seasons ($p < 0.05$, KW test), with the monthly peaks and troughs appearing in April (−50.67 ‰) and September (−93.23 ‰), respectively, which lagged behind those of precipitation isotopes by approximately one month. In contrast, *Q. variabilis* xylem isotope composition remained relatively stable, with no significant seasonal differences ($p > 0.05$; KW test). After entering the rainy season from the dry season, *P. orientalis* and *Q. variabilis* xylem

water isotope signatures showed opposite trends, with the former rapidly decreasing and the latter slowly increasing. After entering the dry season from the rainy season, the isotopic trends of the two species were reversed.

The mean values and ranges of the water isotope signatures showed considerable variation across the precipitation, soil, and xylem samples (Fig. 8). The precipitation isotope signature was significantly different from the 60–80 and 80–100 cm deep soil water isotope signatures ($p < 0.05$, KW test), and the soil water isotope signature at a depth of 0–20 cm was significantly different from that at a depth of 80–100 cm ($p < 0.05$,

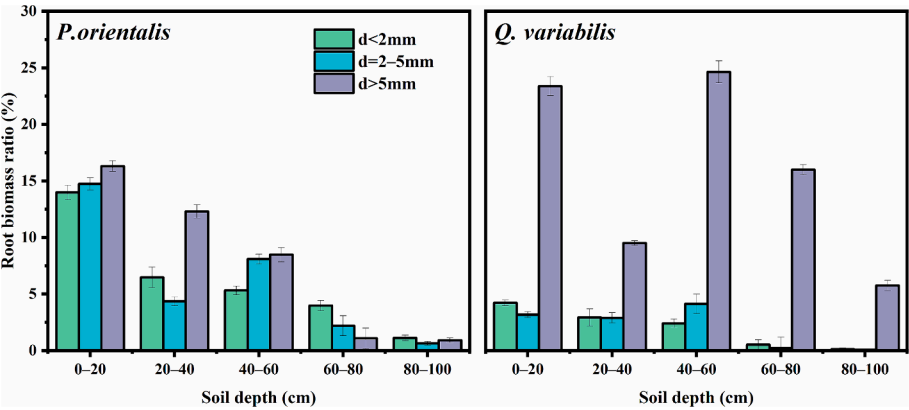


Fig. 4. Distribution of root biomass ratio (%) along the soil profile.

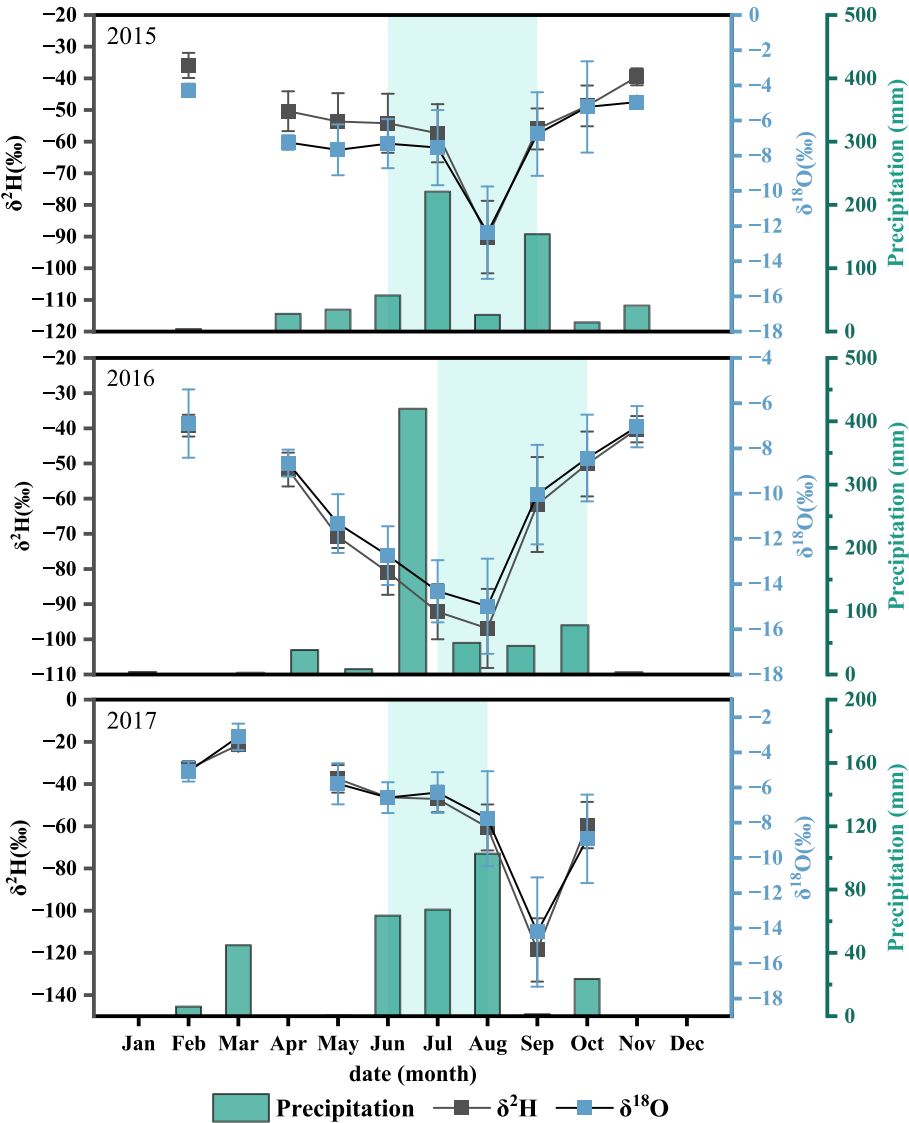


Fig. 5. Monthly variation of precipitation and precipitation isotope signatures over three years in mountainous areas of Beijing. Months with no bars indicate no precipitation during the month and therefore no precipitation isotope data. Light blue areas represent rainy season while white areas represent dry season.

KW test). The range of variability in xylem water isotope signatures was wider for *P. orientalis* than for *Q. variabilis*, suggesting that *P. orientalis* water sources may be further reaching than that of *Q. variabilis*.

3.3. Dual-isotope representation of different water pools

Similar to LMWL, the linear relationship between $\delta^2\text{H}$ and $\delta^{18}\text{O}$ for

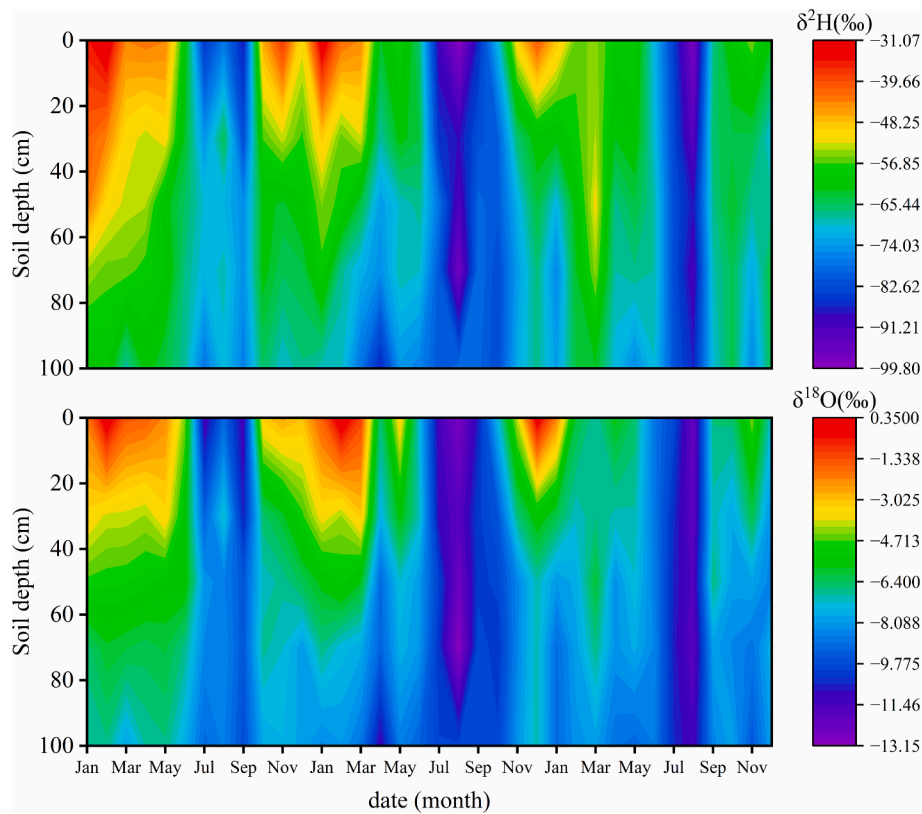


Fig. 6. Variation characteristics of soil water isotope signatures.

soil water and xylem water is defined here as the soil water line (SWL) and xylem water line (XWL), respectively. The vast majority of soil water data points were located at the lower right of the LMWL and gradually converged toward the LMWL with increasing depth (Fig. 9). This indicated that the stable isotopes in soil water at different depths underwent different degrees of evaporative fractionation during the process of recharging soil moisture by precipitation. A comparison of LMWL and SWLs at different soil depths showed that the slopes of SWLs were smaller than those of LMWL, whereas their intercepts were larger than those of LMWL. Most of the xylem water data points were located below the SWL, and the slopes of XWL were lower than those of SWL, whereas their intercepts were greater than those of SWL. The *P. orientalis* xylem water isotopic signature was distributed along both sides of the SWL. In contrast, the *Q. variabilis* xylem isotope signature was relatively stable.

3.4. Variations of lc-excess in different water pools

The spatial and temporal variations in lc-excess for water pools are shown in Fig. 10. The maximum and minimum values of precipitation lc-excess occurred in February 2016 (11.33 ‰) and September 2017 (−15.06 ‰), respectively. As evidence that soil evaporation decreases with depth, soil water lc-excess increases with increasing depth. In the rainy season, the soil water lc-excess in different layers did not differ significantly ($p > 0.05$, KW test). In the dry season, the soil water lc-excess at 0–20 cm was significantly depleted compared to that of other soil layers ($p < 0.05$, KW test), indicating that soil evaporation was greatest within the 0–20 cm depth. During the dry season, soil water and xylem lc-excess of the two species were negative, indicating a deviation from the LWML. *Platycadus orientalis* exhibited lc-excess values near 0 ‰ only during sampling periods in May and June of 2015 and 2016. *Quercus variabilis* consistently showed negative lc-excess throughout the three years. In addition, *Q. variabilis* lc-excess was more negative than that of *P. orientalis* and that of soil water lc-excess.

3.5. MTT and MRT based on the isotope sine curves

The MTT and MRT were calculated based on the ISC to quantify the time trajectory of water transport among the different water pools. We demonstrate the ecohydrological significance of MTT and MRT as represented on the timeline in Fig. 11. On an annual scale, the time period starting with rainfall and ending with the complete mixing of precipitation and bulk soil while the isotope signal fluctuated in a particular soil layer was MTT₁. The time period experienced by water from the root zone was absorbed and transported until transpiration reached MTT₂. The MRT of soil water (MRT_{SW}) was the time that water began to infiltrate until it left the given soil interface (processes such as root uptake, evapotranspiration, and seepage). The MRT of the plant xylem (MRT_{XY}) was the time when water remained in the plant from the time it was absorbed by the roots until it transpired from the leaves.

The isotope sine curve (ISC) parameters and fit statistics for different water pools and the transfer efficiencies among the water pools are shown in Table 1. Compared with precipitation, the amplitude and R^2 of all soil layers water ISC decreased, while the phase increased. The amplitude, R^2 , and offset decreased with increasing depth. These results were sufficient to prove that the precipitation isotope signals gradually weakened during soil infiltration. Comparing plant and soil water, the offset of *P. orientalis* and *Q. variabilis* xylem water ISC was smaller than that of the 0–20 cm soil water ISC and larger than that of the 20–40 cm soil water ISC. The same pattern was observed for the amplitude of *P. orientalis* xylem water ISC. The amplitude of *Q. variabilis* xylem water ISC was lower than that of the 60–80 cm soil water ISC and higher than that of the 80–100 cm soil water ISC.

Our results showed that the transfer efficiency (f) from precipitation to the soil and then to the xylem pool was inconsistent. From 2015 to 2017, the transfer efficiency of this process of precipitation transfer to soil was 65.4 % to 16.7 %, 85.5 % to 13.5 %, and 49.1 % to 17.0 %, respectively. The transfer efficiencies from precipitation to shallow soil were significantly different between years ($p < 0.05$, one-way ANOVA),

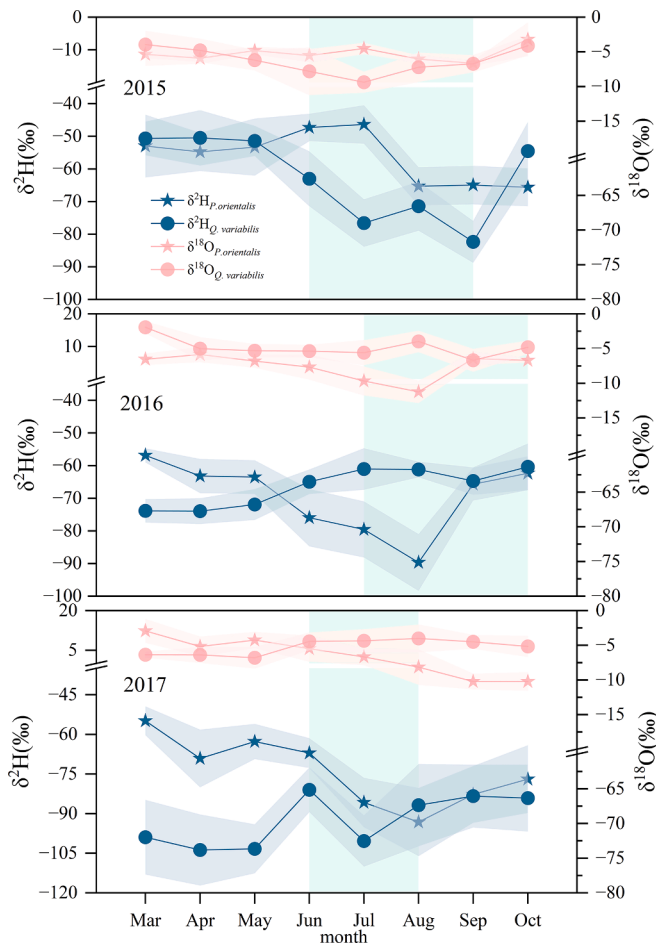


Fig. 7. Variation characteristics of the xylem waters isotope signatures of two species. Xylem water isotope data for November, December, January, and February are missing because most of the tree branches were frozen during the winter months. Light green area represented the rainy season while white areas represented the dry season. Blue and red shading represent standard deviation of the data.

whereas the transfer efficiencies from precipitation to the 60–100 cm soil depth were not significantly different between years ($p > 0.05$, one-way ANOVA). In terms of species, the transfer efficiency from precipitation to *P. orientalis* was greater than that from precipitation to *Q. variabilis* in all cases.

Soil water ISC followed the precipitation trend and gradually sliding to the right (Fig. 12). This was roughly consistent with the infiltration process of precipitation into the soil in space and time, indicating that the hysteresis effect of soil water relative to precipitation increased with increasing soil depth. The *P. orientalis* xylem water ISC fell between 0–20 cm and 20–40 cm soil water ISC, whereas the *Q. variabilis* xylem water ISC fell between 40–60 cm and 60–80 cm soil water ISC.

The MTT and MRT were further calculated based on the ISC to quantify the time trajectory of water transport among different pools (Fig. 13). The MTT₁ first increased and then decreased with increasing soil depth. The minimum value of MTT₁ was 0.5 days for 0–20 cm soil water in 2016 and the maximum value was 15.9 days for 60–80 cm soil water in 2017. In terms of interannual variation, MTT₁ for 0–20 cm and 60–80 cm soil water decreased and for 20–60 cm soil water increased in 2016 relative to 2015. MTT₁ increased for all layers of soil water in 2017 relative to 2015, increased for the 0–20 cm and 60–80 cm soil water layers relative to 2016, and for the 20–60 cm soil water, MTT₁ decreased. MTT₁ had a significantly different trend with soil depth in 2016, compared to the other two years. The MTT₁ of deep soil water was particularly small and close to that of surface soil water. This

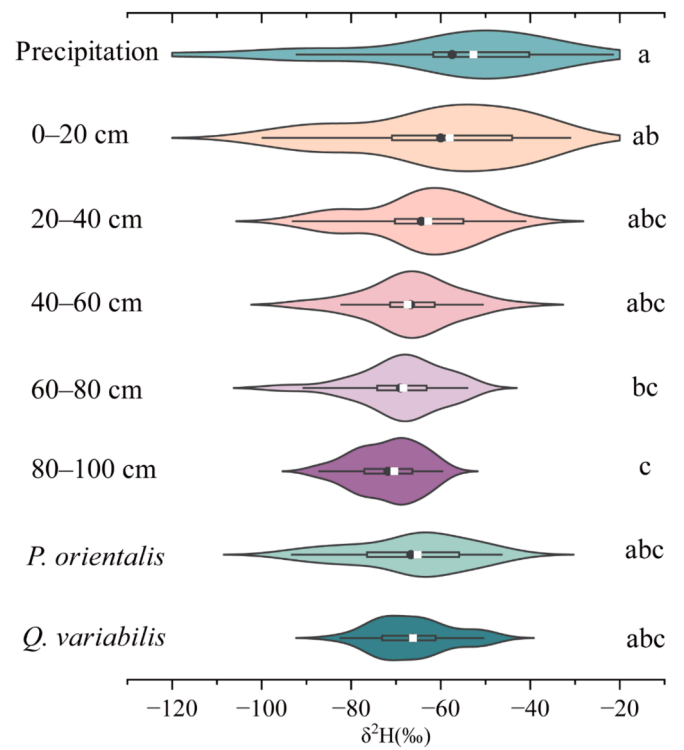


Fig. 8. The violin plot shows the distribution of $\delta^2\text{H}$ in different water pools as estimated by kernel density. The black circle represents the mean, the white square represents the median, the box extent represents the interquartile range, and the whiskers represent the range of outliers. Significance test for pools is indicated by lowercase letters, and groups that do not share letters are significantly different. Significance level is set at $p = 0.05$.

phenomenon may be related to the abundant precipitation in 2016. The high precipitation generated preferential flow in the soil, which formed fast pathways to recharge the deep soil water, and thus the MTT₁ showed a special pattern in 2016.

During the study period, MTT₂ of *P. orientalis* with 0–20 cm soil water remained positive, with the maximum value (4.3d) occurring in 2016, whereas MTT₂ of *P. orientalis* with 20–100 cm soil water was negative, with the minimum value (−13.2d) occurring in 2017 with 60–80 cm soil water (Fig. 14). While MTT₂ of *Q. variabilis* with 20–60 cm varied greatly among years, with the maximum value (11.5d) occurring in 0–20 cm soil water in 2017 and the minimum value (−4.7d) occurring in 20–40 cm soil water in 2016. Overall, MTT₂ of *Q. variabilis* was greater than MTT₂ of *P. orientalis*.

The MRT in different water types is shown in Table 2; MRT_{SW} increased with soil depth. The MRT_{SW} at 0–20 cm was low, which was related to shorter flow paths, indicating that shallow soil water had a faster renewal rate and stronger self-purification ability. The MRT_{SW} at 60–100 cm was high, which was related to longer flow paths, indicating that deep soil stored a large amount of old water and had a slow water renewal cycle. The MRT_{XY} was similar in magnitude to the MRT_{SW} at a depth of 0–20 cm. The MRT_{XY} of *P. orientalis* was not significantly different between the three years, ranging from 53.7 d to 66.0 d. The MRT_{XY} of *Q. variabilis* was the longest in 2016 at 136.8 d, and did not differ much during the other two years.

3.6. Correlation analysis of MTT and MRT with their influencing factors

The MTT₁ was only significantly positively correlated with soil porosity ($p < 0.05$) (Tables 3 and 4); MTT of soil to *P. orientalis* (MTT_{P,o}) was significantly negatively correlated with soil porosity only ($p < 0.05$); MTT of soil to *Q. variabilis* (MTT_{Q,v}) had no correlation with any of

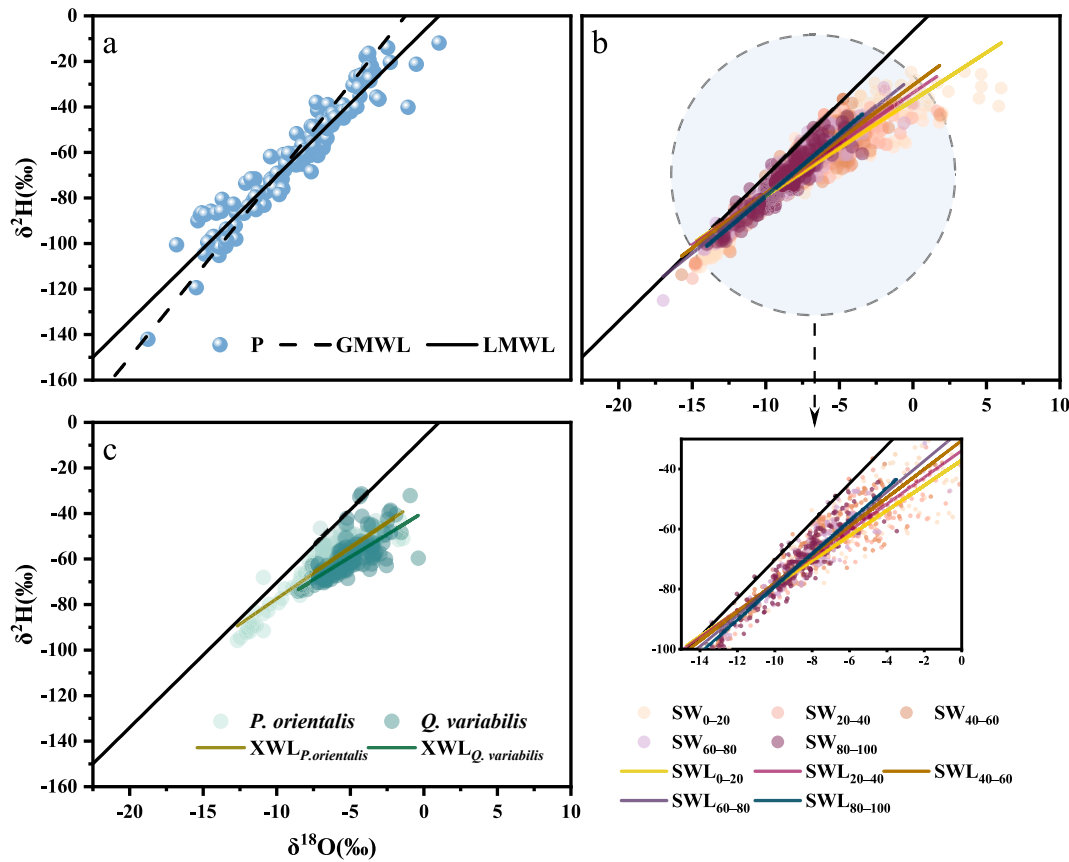


Fig. 9. Dual isotope plots showing isotope data for precipitation (a), soil water (b), xylem water (c). The dashed line indicates the global meteoric water line (GMWL), and the solid line of various colors indicates the fitted line of isotope data. The equations and R^2 values for the fitted lines are as follows, LMWL: $\delta^2\text{H} = 6.37 \times \delta^{18}\text{O} - 6.61$, $R^2 = 0.917$; SW_{0-20} : $\delta^2\text{H} = 4.19 \times \delta^{18}\text{O} - 37.09$, $R^2 = 0.825$; SW_{20-40} : $\delta^2\text{H} = 4.44 \times \delta^{18}\text{O} - 33.87$, $R^2 = 0.828$; SW_{40-60} : $\delta^2\text{H} = 4.77 \times \delta^{18}\text{O} - 30.50$, $R^2 = 0.826$; SW_{60-80} : $\delta^2\text{H} = 5.15 \times \delta^{18}\text{O} - 26.98$, $R^2 = 0.872$; SW_{80-100} : $\delta^2\text{H} = 5.48 \times \delta^{18}\text{O} - 24.45$, $R^2 = 0.855$; $\text{XWL}_{P.orientalis}$: $\delta^2\text{H} = 4.45 \times \delta^{18}\text{O} - 32.90$, $R^2 = 0.835$; $\text{XWL}_{Q.variabilis}$: $\delta^2\text{H} = 3.99 \times \delta^{18}\text{O} - 39.31$, $R^2 = 0.510$.

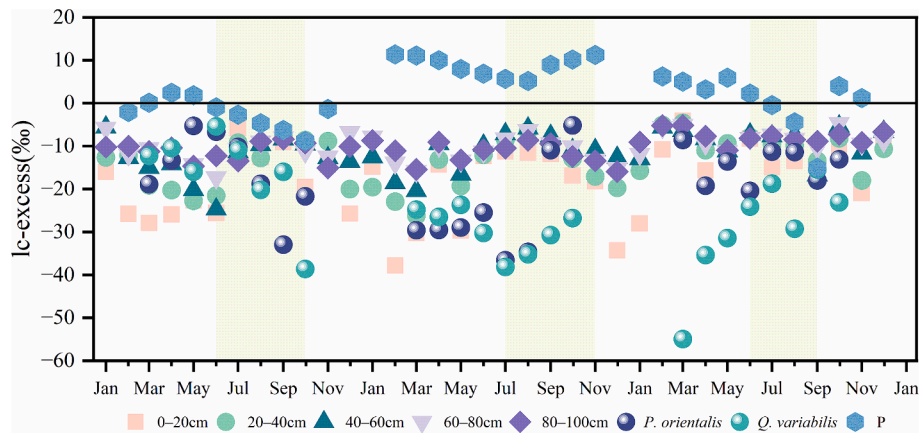


Fig. 10. Temporal variations of lc-excess in precipitation (P), soil water in all layers, and xylem water of *P. orientalis* and *Q. variabilis*. Light cyan area represented the rainy season while white areas represented the dry season. For visual clarity, both the mean and standard error are not shown.

the factors; MRT_{SW} was significantly negatively correlated with the root biomass distribution of *P. orientalis* ($p < 0.01$), and significantly negatively correlated with the fine and medium root biomass distributions of *Q. variabilis* ($p < 0.01$); at the same time, MRT_{SW} was significantly negatively correlated with the clay content in the soil ($p < 0.05$), significantly positively correlated with soil bulk weight ($p < 0.01$), and significantly negatively correlated with SWC ($p < 0.05$). For both species, MRT_{XY} was significantly and positively correlated with height and

DBH ($p < 0.01$), and was not correlated with either VPD or solar radiation (Table 5).

4. Discussion

4.1. Transfer efficiency indicating connectivity between water pools

Assessing the age distribution at interfaces is often challenging when

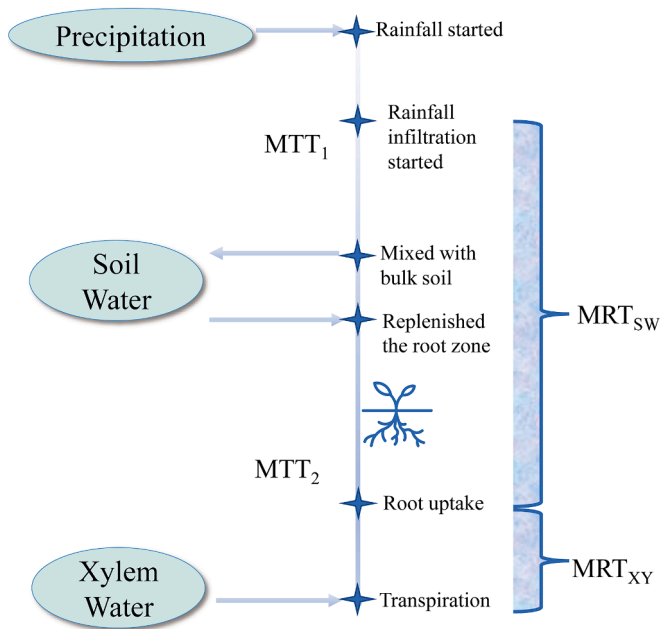


Fig. 11. Ecohydrological significance of MTT and MRT on the timeline. The four-pointed star indicates important time points in the water transport process.

considering the transboundary and feedback effects of water transport between water pools (Staudinger, et al., 2019). Therefore, it is necessary to study hydrological cycle processes in combination with dynamics within water pools and connectivity among water pools.

On the time scale of a single rainfall event, the pattern of change in ΔSWC showed that shallow soil was more sensitive to precipitation, indicating greater connectivity between precipitation and shallow soil water. Beneath the maximum evaporation penetration depth, infiltrated precipitation water mixed with previously stored water over seasons and years so that the soil water isotope signature represents the long-term

average of previous precipitation events that have recharged these soils. At the annual scale, we observed no significant isotopic differences between precipitation and 0–60 cm soil water, and significant isotopic differences between precipitation and 60–100 cm soil water (Fig. 8). Meanwhile, the DF results showed that water transfer efficiency gradually decreased with increasing depth, and the connectivity between soil water and precipitation gradually weakened. However, the lc-excess showed the opposite pattern. In the dry season, the soil water lc-excess decreased with increasing depth, indicating that the connectivity between soil water and precipitation gradually increased with increasing depth. These two indicators yielded different results. In terms of tree species, the transfer efficiency from precipitation to *P. orientalis* was higher than that from precipitation to *Q. variabilis*, whereas *Q. variabilis* lc-excess was more negative than *P. orientalis*. The two indicators remained consistent in characterizing the connectivity between precipitation and xylem water, suggesting greater connectivity between precipitation and *P. orientalis*. Notably, we found that during the rainy season, the soil water lc-excess hardly varied with depth. Therefore, we can conclude that in the dry season, soil evaporation was strong, which masked the signal of isotope transmission, whereas in the rainy season, the signal of isotope transmission was partially revealed when soil evaporation was relatively weak owing to heavy rainfall (Ju et al., 2024). It was difficult for soil evaporation to affect water transfer within the trees. Therefore, DF, which is not disturbed by soil evaporation, reflects the connectivity among different water pools better and more clearly than lc-excess.

4.2. The spatiotemporal heterogeneity of MTT and MRT

Through our analysis of precipitation, soil, and plant water isotopes, we illustrated the propagation of the water isotope signal as it traveled through SPAC, while also examining its temporal variability. Precipitation took a longer path and more time to enter deep soil than shallow soil. Therefore, shallow soil water is more sensitive to precipitation, with a more rapid renewal rate, and soil water age increases with soil depth. This was consistent with the variations in MTT_1 and MRT_{SW} (Fig. 10, Table 3). For example, in years with high annual rainfall, MTT_1 and

Table 1

Sine parameters (amplitude, phase, and offset), fit statistics (R^2 , p) for different water pools. Transfer efficiency (f) among water pools.

Year	Different water pools	$\delta^2\text{H}$					
		amplitude	φ	offset	R^2	p	f
2015	Precipitation	33.763	−0.884	−56.889	0.690	0.000	100 %
	0–20 cm	22.060	−0.870	−53.779	0.485	0.000	65.3 %
	20–40 cm	13.108	−0.838	−57.567	0.360	0.000	38.8 %
	40–60 cm	12.327	−0.799	−60.656	0.313	0.000	36.5 %
	60–80 cm	8.158	−0.657	−62.716	0.218	0.000	24.2 %
	80–100 cm	5.625	−0.701	−67.415	0.086	0.023	16.7 %
	<i>P. orientalis</i>	15.070	−0.832	−56.735	0.455	0.000	44.6 %
	<i>Q. variabilis</i>	8.096	−0.736	−56.922	0.268	0.000	24.0 %
2016	Precipitation	33.010	−0.974	−57.052	0.693	0.000	100 %
	0–20 cm	28.223	−0.965	−62.344	0.680	0.000	85.5 %
	20–40 cm	16.673	−0.723	−68.301	0.472	0.000	50.5 %
	40–60 cm	11.631	−0.753	−71.203	0.318	0.000	35.2 %
	60–80 cm	9.812	−0.765	−74.094	0.260	0.000	29.7 %
	80–100 cm	4.464	−0.961	−76.372	0.082	0.039	13.5 %
	<i>P. orientalis</i>	18.653	−0.891	−67.595	0.603	0.000	56.5 %
	<i>Q. variabilis</i>	6.379	−0.804	−67.585	0.246	0.000	19.3 %
2017	Precipitation	32.997	−1.058	−59.316	0.479	0.000	100 %
	0–20 cm	16.192	−1.035	−64.119	0.426	0.000	49.1 %
	20–40 cm	10.985	−0.876	−66.838	0.323	0.000	33.3 %
	40–60 cm	7.739	−0.862	−68.331	0.185	0.003	23.5 %
	60–80 cm	6.989	−0.784	−70.019	0.133	0.014	21.2 %
	80–100 cm	5.611	−1.003	−71.446	0.090	0.043	17.0 %
	<i>P. orientalis</i>	11.893	−1.105	−66.612	0.384	0.000	36.0 %
	<i>Q. variabilis</i>	5.661	−0.837	−64.656	0.235	0.001	16.9 %

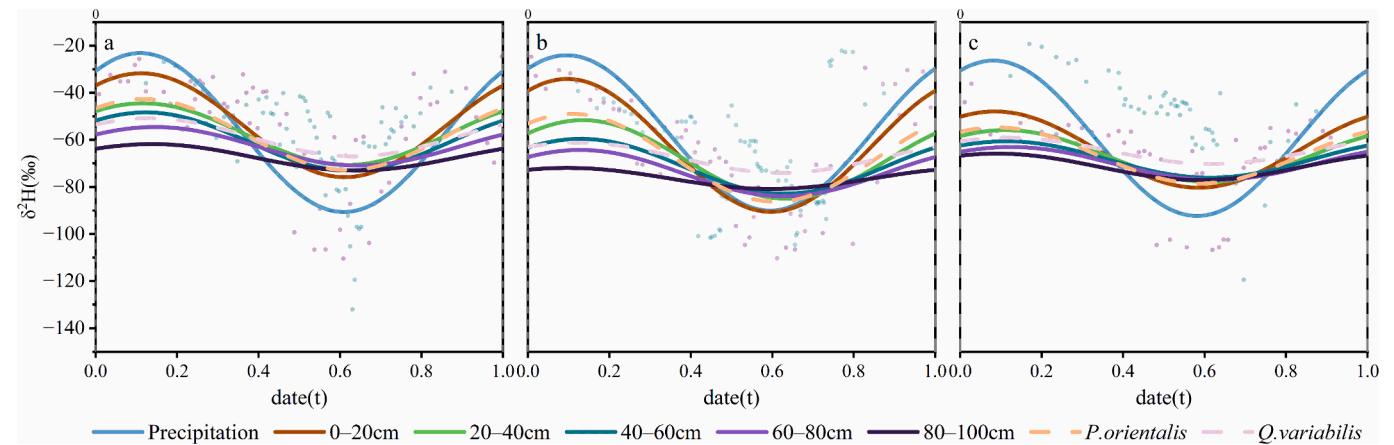


Fig. 12. Isotope sine curve (ISC) plots for different pools in 2015 (a), 2016 (b), and 2017 (c). The light green and light purple dots represent the precipitation and soil water datasets, respectively. The regression parameters, R^2 and p -values for each fitted line have been shown in Table 1.

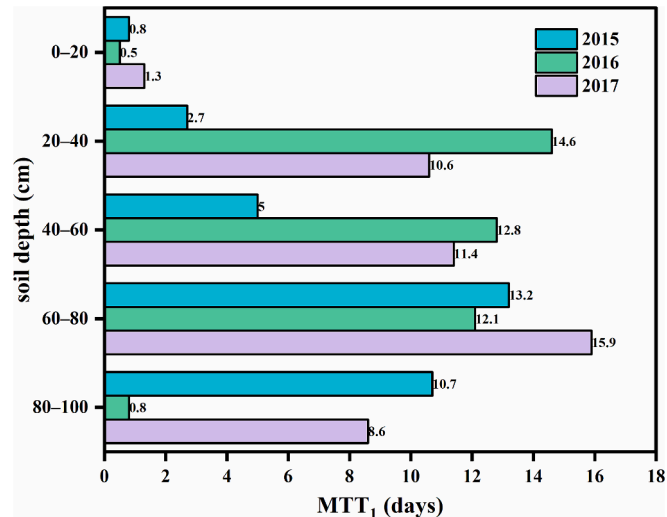


Fig. 13. Mean transmission time of precipitation to soil (MTT_1) of different soil layers during study period.

MRT_{SW} of shallow soil decreased significantly, while MTT_1 and MRT_{SW} of deep soil changed little. This reflects the form of soil water movement in the vertical direction; overall, it was mainly piston flow, although preferential flow was also present (Sprenger et al., 2019a). When soil water infiltrates in the form of piston flow, the water advances layer-by-

layer (Liu et al., 2022a). Deep soil water mainly relies on heavy precipitation replenishment, with MRT_{SW} increasing with depth. At the same time, fast pathways were formed by the connection of large pores in the soil, which allowed part of the precipitation to bypass shallow soil and directly replenish deeper soil in the form of preferential flow (Ju et al., 2024). This also explains why MTT_1 decreased after an initial rise, following an increased soil depth. The larger values of MRT_{SW} in the middle and deep soil layers indicated that soil water in the recent season not only contained the water replenished by precipitation in the recent season but also water replenished by precipitation in different seasons in the previous period.

Because the plant xylem acted as a pool for multiple soil water mixing events, the contribution of soil water from each layer was not uniform. Our results showed that *P. orientalis* xylem water ISC fell between 0–20 cm and 20–40 cm soil water ISC, whereas *Q. variabilis* xylem

Table 2
Mean residence time (MRT) in different water pools.

Water pools	MRT (d)		
	2015	2016	2017
SW _{0–20cm}	67.3	35.2	103.1
SW _{20–40cm}	137.9	99.3	164.5
SW _{40–60cm}	148.1	154.3	240.8
SW _{60–80cm}	233.3	186.6	268.0
SW _{80–100cm}	343.8	325.6	336.6
<i>P. orientalis</i>	62.1	66.0	53.7
<i>Q. variabilis</i>	73.8	136.8	73.2

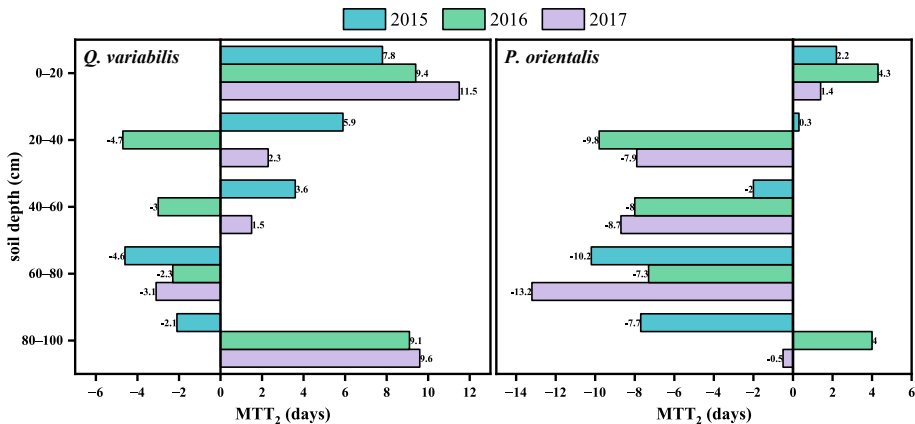


Fig. 14. Mean transmission time of soil to trees (MTT_2) of two species during study period.

Table 3Correlations (Spearman coefficients) of time lag with soil physicochemical properties and *P. orientalis* root distribution.

	<i>P. orientalis</i> root distribution			Soil separates			Soil bulk density	Soil porosity	SWC
	thin	middle	thick	clay	silt	sand			
MTT ₁	−0.349	−0.328	−0.349	−0.328	−0.317	0.317	−0.098	0.551*	−0.222
MTT _{P.o}	0.295	0.273	0.295	0.251	0.360	−0.360	0.164	−0.556*	0.264
MRT _{SW}	−0.927**	−0.840**	−0.927**	−0.535*	−0.120	0.120	0.644**	−0.306	−0.543*

Table 4Correlations (Spearman coefficients) of time lag with soil physicochemical properties and *Q. variabilis* root distribution.

	<i>Q. variabilis</i> root distribution			Soil separates			Soil bulk density	Soil porosity	SWC
	thin	middle	thick	clay	silt	sand			
MTT ₁	−0.349	−0.093	0.038	−0.328	−0.317	0.317	−0.098	0.551*	−0.222
MTT _{Q.v}	0.164	0.055	0.000	0.142	0.131	−0.131	0.022	−0.207	0.418
MRT _{SW}	−0.927**	−0.655**	−0.491	−0.535*	−0.120	0.120	0.644**	−0.306	−0.543*

Table 5

Correlations (Spearman coefficients) of MRT with tree morphological traits and environmental factors.

	Height	DBH	VPD	R
MRT _{P.o}	0.857**	0.698**	−0.111	0.251
MRT _{Q.v}	0.786**	0.648**	−0.149	0.192

**, $P < 0.01$; *, $P < 0.05$.

water ISC fell between 40–60 cm and 60–80 cm soil water ISC. The positive and negative MTT₂ values of *P. orientalis* and *Q. variabilis* were also consistent with this soil distribution pattern. Therefore, it could be deduced that shallow soil water contributes more to the uptake of *P. orientalis* and deep soil water contributes more to the uptake of *Q. variabilis*. This result is similar to the conclusions of Liu, et al., 2023. *Platycadus orientalis* has a well-developed shallow root system that is sensitive to precipitation, tends to absorb shallow soil water in the rainy season, and obtains water mainly from deep soil with relatively high SWC in the dry season because only specific moist conditions can activate the water absorption capacity of the plant root system (Liu et al., 2019). In contrast, the *Q. variabilis* root system was not sensitive to precipitation (Jia et al., 2022) and mainly utilized the more stable deep soil water in both dry and rainy seasons.

During the drought season, when the water supplied to the upper soil layer was reduced, both *P. orientalis* and *Q. variabilis* used soil water from deep soil. Given the increased MRT of deep soil water (Table 3), the two species differed in their ability to access deep soil water during drought, thus explaining the greater MRT_{XY}; i.e., the greater proportion of old water was absorbed by *Q. variabilis* compared with *P. orientalis*. These results are similar to those reported in recent studies. For example, the MRT within hemlock ranged from 20 to 100 days, whereas for beech, it ranged from 5 to 50 days (Knighton et al., 2020). The estimated WTT of five tree species in a dry forest ecosystem ranged from 11 to 22 days, whereas the MRT for deciduous trees was approximately twice as high as that for evergreen trees because transpiration is weaker in deciduous trees than in evergreen species during the non-growing season (Graefe et al., 2019). Owing to the difference in hydraulic conductivity, the difference in MRT between broadleaf and needle species was also obvious, and the MRT of the needle species *Abies fabri* was significantly higher than that of the broadleaf species *Fagus longipetiolata* (Magh et al., 2020).

4.3. Influencing factors of MTT and MRT and underlying mechanisms

Soil structure is three-dimensional. As a result, both vertical and lateral flow rates vary, leading to lateral isotopic variations that can be as significant as vertical variations (Penna and van Meerveld, 2019;

Sprenger and Allen, 2020), which complicates the identification of the time trajectories of water travel. Trees are not simple transport vessels (Evaristo et al., 2019), with their ecological and biological characteristics determining their water use strategy (Hochberg et al., 2018). This, in turn, affects the water transmission rates within trees. At the same time, the exchange between the transpiration stream and storage water affects WRT within trees (Seeger and Weiler, 2021). Thus, the factors influencing water travel time trajectories and underlying mechanisms are complex.

Our results showed that differences in root distribution (vertical variability) and soil pores containing water percolating at different rates (horizontal variability) positively and negatively influenced water movement and cross-interfacial processes to varying degrees. This was corroborated by previous findings (Barbata, 2019; Bowers et al., 2020; Sprenger, et al., 2019a). Sprenger, et al. (2019a) emphasized the extensive duration required for mixing and equilibration across pores, even in the absence of a tension influence. Consequently, isotopic heterogeneities in soils arise not only as an inevitable result of tension variations, but also due to disparities in the flow of water through different pores (Sprenger and Allen, 2020). In smaller pores, which are less conductive, water flows slower and has a longer residence time. Owing to the mechanism of root growth toward water (Bao et al., 2014), the root system is biased to occupy smaller pores. At the same time, plants reduce the conductivity of the soil pores around the root system through compaction and mucus production (York et al., 2016), resulting in a further reduction in porosity; thus, the retained water absorbed by the root system has the potential to span seasons. Allen et al. (2019) demonstrated that broadleaf trees and conifers utilize water with distinct isotopic compositions despite having similar shallow root distributions. This suggests that the two tree types access water from different pore domains. A hypothetical explanation for this phenomenon is that broadleaf trees possess finer roots with smaller diameters, allowing them to occupy narrower soil pores than conifers (Sprenger and Allen, 2020).

Our results showed a significant negative correlation between MRT_{SW} and soil clay content ($p < 0.05$). These results are similar to those reported by Sprenger et al. (2019a), who emphasized that water held in smaller soil pores of the clayey fraction appears to be considerably younger than water in the pores of the sandy soil texture within the same soil profile. The MRT_{SW} has a significant impact on plant growth and can reveal the effects of environmental changes on water cycling. Notably, a longer MRT_{SW} is not always beneficial. Excessive MRT_{SW} can impede water infiltration, leading to root anoxia that hinders plant growth. Conversely, when MRT_{SW} is too short, insufficient recharge may result in water deficits (Ma et al., 2019). In an entire watershed, a long MRT_{SW} can affect the entire hydrological cycle (Jeelani et al., 2013). Therefore, controlling MRT_{SW} and optimizing plant-soil hydrological processes

require further research.

We observed that MRT_{XY} was influenced by tree height and DBH. Therefore, incorporating tree storage and internal mixing into an eco-hydrological framework facilitates an accurate assessment of plant water use strategies in critical zones (Knighton et al., 2020). The MRT_{XY} can account for tree water storage and the extent of exchange within tree tissues (Meinzer et al., 2006). This is related to the hydraulic capacitance of sapwood, which facilitates water exchange between sapwood tissues and xylem conduit lumens and has the potential to influence apparent hydraulic conductance by mitigating changes in the soil-to-leaf water potential gradient (Gaines et al., 2016). Water stored as a result of sapwood capacitance can account for a significant portion of the stem mass; however, only a small fraction of the stored water contributes to the transpiration flux (Vargas et al., 2017).

5. Conclusions

We demonstrated the propagation paths of water isotope signals and quantified the transmission and residence times. By combining the dynamics within the pools with connectivity between the pools, we restored the temporal trajectory of water traveling through SPAC. Our study suggests that the damping factor (DF), which is not affected by soil evaporation, is a better indicator of connectivity among different water pools than ΔSWC and lc -excess. The spatiotemporal pattern of MTT_1 assessed the age distribution of water at the interface of different soil layers. MTT_2 indicated the tendency of plant water uptake, with *P. orientalis* being more sensitive to water variations. MRT was influenced by a combination of soil physicochemical properties, root distribution, and tree morphological traits, so that the water actually absorbed by plants can span seasons. Our findings provide new insights into the cross-boundary and feedback effects of water movement in different water pools.

CRedit authorship contribution statement

Ziqi Liu: Writing – original draft, Methodology, Data curation, Conceptualization. **Zihe Liu:** Writing – review & editing, Methodology. **Guodong Jia:** Writing – review & editing, Visualization, Funding acquisition. **Ziqiang Liu:** Investigation. **Xinxiao Yu:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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References

- Allen, S.T., Kirchner, J.W., Goldsmith, G.R., J.G.R.L., 2018. Predicting spatial patterns in precipitation isotope (δ^2H and $\delta^{18}O$) seasonality using sinusoidal isoscapes. *Geophys. Res. Lett.* 45 (10), 4859–4868.
- Allen, S.T., Kirchner, J.W., Braun, S., Siegwolf, R.T.W., Goldsmith, G.R., 2019. Seasonal origins of soil water used by trees. *Hydrol. Earth Syst. Sci.* 23 (2), 1199–1210. <https://doi.org/10.5194/hess-23-1199-2019>.

- Bao, Y., et al., 2014. Plant roots use a patterning mechanism to position lateral root branches toward available water. *PNAS* 111 (25), 9319–9324. <https://doi.org/10.1073/pnas.1400966111>.
- Barbeta, A., 2019. Unexplained hydrogen isotope offsets complicate the identification and quantification of tree water sources in a riparian forest. *Hydrol. Earth Syst. Sci.* 23 (4), 2129–2146. <https://doi.org/10.5194/hess-23-2129-2019>.
- Bowers, W.H., Mercer, J.J., Pleasants, M.S., Williams, D.G., 2020. A combination of soil water extraction methods quantifies the isotopic mixing of waters held at separate tensions in soil. *Hydrol. Earth Syst. Sci.* 24 (8), 4045–4060. <https://doi.org/10.5194/hess-24-4045-2020>.
- Brinkmann, N., 2018. Employing stable isotopes to determine the residence times of soil water and the temporal origin of water taken up by *Fagus sylvatica* and *Picea abies* in a temperate forest. *New Phytol.* 219 (4), 1300–1313. <https://doi.org/10.1111/nph.15255>.
- Correa, A., 2020. Modelling non-stationary water ages in a tropical rainforest: A preliminary spatially distributed assessment. *Hydrol. Process.* 34 (25), 4776–4793.
- Craig, H., 1961. Isotopic Variations in Meteoric Waters. *Science* 133 (3465), 1702–1703. <https://doi.org/10.1126/science.133.3465.1702>.
- Dai, J.J., et al., 2020. Variation of the stable isotopes of water in the soil-plant-atmosphere continuum of a woodland in the East Asian monsoon region. *J. Hydrol.* 589, 125199. <https://doi.org/10.1016/j.jhydrol.2020.125199>.
- Evaristo, J., 2019. Characterizing the Fluxes and Age Distribution of Soil Water, Plant Water and Deep Percolation in a Model Tropical Ecosystem. *Water Resour. Res.* 55 (4), 3307–3327. <https://doi.org/10.1029/2018wr023265>.
- Gaines, K.P., Meinzer, F.C., Duffy, C.J., Thomas, E.M., Eissenstat, D.M., 2016. Rapid tree water transport and residence times in a Pennsylvania catchment. *Ecohydrology* 9 (8), 1554–1565. <https://doi.org/10.1002/eco.1747>.
- Geris, J., Tetzlaff, D., McDonnell, J.J., Soulsby, C., 2017. Spatial and temporal patterns of soil water storage and vegetation water use in humid northern catchments. *Sci. Total Environ.* 595, 486–493. <https://doi.org/10.1016/j.scitotenv.2017.03.275>.
- Goldsmith, G.R., 2019. Spatial variation in throughfall, soil, and plant water isotopes in a temperate forest. *Ecohydrology* 12 (2), e2059.
- Graefe, S., Fang, D.M., Butz, P., 2019. Water residence times in trees of a neotropical dry forest. *Trees-Structure and Function* 33 (4), 1225–1231. <https://doi.org/10.1007/s00468-019-01849-y>.
- Hervé-Fernández, P., 2016. Assessing the ‘two water worlds’ hypothesis and water sources for native and exotic evergreen species in south-central Chile. *Hydrol. Process.* 30 (23), 4227–4241.
- Hochberg, U., Rockwell, F.E., Holbrook, N.M., Cochard, H.J.T.i.p.s., 2018. Iso/anisohydry: a plant–environment interaction rather than a simple hydraulic trait. *Trends in plant science*.
- Huang, M., Hilderman, J.N., Barbour, L.J.J.O.H., 2015. Transport of stable isotopes of water and sulphate within reclaimed oil sands saline–sodic mine overburden. *J. Hydrol.* 529, 1550–1561.
- Jeelani, G., Kumar, U.S., Kumar, B.J.J.O.H., 2013. Variation of $\delta^{18}O$ and δD in precipitation and stream waters across the Kashmir Himalaya (India) to distinguish and estimate the seasonal sources of stream flow. *J. Hydrol.* 481, 157–165.
- Jia, G.D., Nehemy, M.F., Chen, L.X., Yu, X.X., Liu, Z.Q., 2022. Ephemeral connectivity between trees and groundwater in a temperate forest in China. *J. Hydrol.* 610, 127887. <https://doi.org/10.1016/j.jhydrol.2022.127887>.
- Ju, X., 2024. Impacts of the soil pore structure on infiltration characteristics at the profile scale in the red soil region. *Soil Tillage Res.* 236, 105922.
- Juhlke, T.R., 2021. Temporal offset between precipitation and water uptake of Mediterranean pine trees varies with elevation and season. *Sci. Total Environ.* 755, 142539. <https://doi.org/10.1016/j.scitotenv.2020.142539>.
- Kabeya, N., Katsuyama, M., Kawasaki, M., Ohte, N., Sugimoto, A., 2007. Estimation of mean residence times of subsurface waters using seasonal variation in deuterium excess in a small headwater catchment in Japan. *Hydrol. Process.* 21 (3), 308–322. <https://doi.org/10.1002/hyp.6231>.
- Knighton, J., 2020. Using isotopes to incorporate tree water storage and mixing dynamics into a distributed ecohydrologic modelling framework. *Ecohydrology* 13 (3), e2201. <https://doi.org/10.1002/eco.2201>.
- Liu, H., 2022. Estimation of variability in soil water content in a forested critical-zone experimental catchment in Eastern China. *J. Contam. Hydrol.* 248, 104022. <https://doi.org/10.1016/j.jconhyd.2022.104022>.
- Liu, Q., 2022. Characterizing the spatiotemporal dynamics of shallow soil water stable isotopic compositions on a karst hillslope in Southwestern China. *J. Hydrol.* 610, 127964. <https://doi.org/10.1016/j.jhydrol.2022.127964>.
- Liu, Z., 2023. Increased uptake of deep soil water promotes drought resistance in mixed forests. *Plant Cell Environ.* 46 (11), 3218–3228. <https://doi.org/10.1111/pce.14642>.
- Liu, Z.Q., Yu, X.X., Jia, G.D., 2019. Water uptake by coniferous and broad-leaved forest in a rocky mountainous area of northern China. *Agric. For. Meteorol.* 265, 381–389. <https://doi.org/10.1016/j.agrformet.2018.11.036>.
- Luo, Z.D., Nie, Y.P., Ding, Y.L., Chen, H.S., 2021. Replenishment and mean residence time of root-zone water for woody plants growing on rocky outcrops in a subtropical karst critical zone. *J. Hydrol.* 603, 127136. <https://doi.org/10.1016/j.jhydrol.2021.127136>.
- Ma, J., Song, W., Wu, J., Liu, Z., Wei, Z., 2019. Identifying the mean residence time of soil water for different vegetation types in a water source area of the Yuanjiang Terrace, southwestern China. *Isot. Environ. Health Stud.* 55 (3), 272–289. <https://doi.org/10.1080/10256016.2019.1601090>.
- Magh, R.K., 2020. Competition for water rather than facilitation in mixed beech-fir forests after drying-wetting cycle. *J. Hydrol.* 587, 124944. <https://doi.org/10.1016/j.jhydrol.2020.124944>.

- Martin, J., Looker, N., Hoylman, Z., Jencso, K., Hu, J., 2018. Differential use of winter precipitation by upper and lower elevation Douglas fir in the Northern Rockies. *Glob. Chang. Biol.* 24 (12), 5607–5621. <https://doi.org/10.1111/gcb.14435>.
- Meinzer, F.C., 2006. Dynamics of water transport and storage in conifers studied with deuterium and heat tracing techniques. *Plant Cell Environ.* 29 (1), 105–114. <https://doi.org/10.1111/j.1365-3040.2005.01404.x>.
- Mennekes, D., Rinderer, M., Seeger, S., Orlowski, N., 2021. Ecohydrological travel times derived from in situ stable water isotope measurements in trees during a semi-controlled pot experiment. *Hydrol. Earth Syst. Sci.* 25 (8), 4513–4530. <https://doi.org/10.5194/hess-25-4513-2021>.
- Miguez-Macho, G., Fan, Y., 2021. Spatiotemporal origin of soil water taken up by vegetation. *Nature* 598 (7882), 624–628. <https://doi.org/10.1038/s41586-021-03958-6>.
- Penna, D., van Meerveld, H.J., 2019. Spatial variability in the isotopic composition of water in small catchments and its effect on hydrograph separation. *Wiley Interdiscipl. Rev. Water* 6 (5), e1367. <https://doi.org/10.1002/wat2.1367>.
- Reddy, M.M., Schuster, P., Kendall, C., Reddy, M.B.J.H.P., 2006. Characterization of surface and ground water $\delta^{18}\text{O}$ seasonal variation and its use for estimating groundwater residence times. *Hydrol. Process.* 20 (8).
- Rempe, D.M., Dietrich, W.E., 2018. Direct observations of rock moisture, a hidden component of the hydrologic cycle. *PNAS* 115 (11), 2664–2669. <https://doi.org/10.1073/pnas.1800141115>.
- Renée Brooks, J., Barnard, H.R., Coulombe, R., McDonnell, J.J.J.N.G., 2010. Ecohydrologic separation of water between trees and streams in a Mediterranean climate. *Nat. Geosci.* 3 (2), 100–104.
- Rodgers, P., Soulsby, C., Waldron, S., 2005. Stable isotope tracers as diagnostic tools in upscaling flow path understanding and residence time estimates in a mountainous mesoscale catchment. *Hydrol. Process.* 19 (11), 2291–2307. <https://doi.org/10.1002/hyp.5677>.
- Seeger, S., Weiler, M., 2021. Temporal dynamics of tree xylem water isotopes: in situ monitoring and modeling. *Biogeosciences* 18 (15), 4603–4627. <https://doi.org/10.5194/bg-18-4603-2021>.
- Sprenger, M., 2018. Storage, mixing, and fluxes of water in the critical zone across northern environments inferred by stable isotopes of soil water. *Hydrol. Process.* 32 (12), 1720–1737. <https://doi.org/10.1002/hyp.13135>.
- Sprenger, M., 2019. The demographics of water: a review of water ages in the critical zone. *Rev. Geophys.* 57 (3), 800–834. <https://doi.org/10.1029/2018rg000633>.
- Sprenger, M., Allen, S.T., 2020. What Ecohydrologic Separation Is and Where We Can Go With It. *Water Resour. Res.* 56(7): e2020WR027238. [10.1029/2020WR027238](https://doi.org/10.1029/2020WR027238).
- Sprenger, M., Volkmann, T.H.M., Blume, T., Weiler, M., 2015. Estimating flow and transport parameters in the unsaturated zone with pore water stable isotopes. *Hydrol. Earth Syst. Sci.* 19 (6), 2617–2635. <https://doi.org/10.5194/hess-19-2617-2015>.
- Sprenger, M., Seeger, S., Blume, T., Weiler, M., 2016. Travel times in the vadose zone: Variability in space and time. *Water Resour. Res.* 52 (8), 5727–5754. <https://doi.org/10.1002/2015wr018077>.
- Sprenger, M., Tetzlaff, D., Soulsby, C.J.H., Sciences, E.S., 2017. Soil water stable isotopes reveal evaporation dynamics at the soil–plant–atmosphere interface of the critical zone. *J. Hydrol.* 21 (7), 3839–3858.
- Sprenger, M., Llorens, P., Cayuela, C., Gallart, F., Latron, J., 2019. Mechanisms of consistently disjunct soil water pools over (pore) space and time. *Hydrol. Earth Syst. Sci.* 23 (6), 2751–2762. <https://doi.org/10.5194/hess-23-2751-2019>.
- Staudinger, M., 2019. Your work is my boundary condition! Challenges and approaches for a closer collaboration between hydrologists and hydrogeologists. *J. Hydrol.* 571, 235–243. <https://doi.org/10.1016/j.jhydrol.2019.01.058>.
- Tetzlaff, D., 2021. Stable isotopes of water reveal differences in plant–soil water relationships across northern environments. *Hydrol. Process.* 35 (1), e14023.
- Vargas, A.I., Schaffer, B., Yuhong, L., Sternberg, L., 2017. Testing plant use of mobile vs immobile soil water sources using stable isotope experiments. *New Phytol.* 215 (2), 582–594. <https://doi.org/10.1111/nph.14616>.
- von Freyberg, J., Scott T.Grossiord, CharlotteDawson, Todd E. %J Methods in Ecology, Evolution, 2020. Plant and root-zone water isotopes are difficult to measure, explain, and predict: Some practical recommendations for determining plant water sources. *Methods in Ecology Evolution*, 11(11).
- Wan, L., 2023. What determines the time lags of sap flux with solar radiation and vapor pressure deficit? *Agric. For. Meteorol.* 333, 109414.
- Wenping, D., 2021. Long-term study on the seasonal water uptake of *Platycladus orientalis* in the Beijing mountain area, northern China. *Agric. For. Meteorol.* 307, 108531.
- Wu, J.E., 2022. Plant hydrological niches become narrow but stable as the complexity of interspecific competition increases. *Agric. For. Meteorol.* 320, 108953. <https://doi.org/10.1016/j.agrformet.2022.108953>.
- York, L.M., Carminati, A., Mooney, S.J., Ritz, K., Bennett, M.J., 2016. The holistic rhizosphere: integrating zones, processes, and semantics in the soil influenced by roots. *J. Exp. Bot.* 67 (12), 3629–3643. <https://doi.org/10.1093/jxb/erw108>.
- Zhao, Y., Wang, L., 2021. Insights into the isotopic mismatch between bulk soil water and Koidz trunk water from root water stable isotope measurements. *Hydrol. Earth Syst. Sci.* 25 (7), 3975–3989. <https://doi.org/10.5194/hess-25-3975-2021>.